

POLITECNICO
MILANO 1863
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

RF Systems

Noise

Noise generated by a resistor at T (Kelvin) (Thermal noise)

The **Available Noise Power** produced by a resistor is given by:

$$P_N = kT\Delta f$$

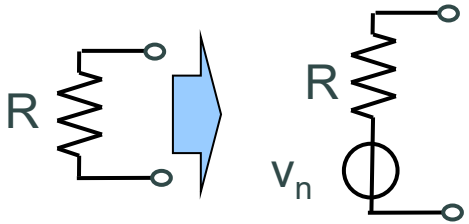
$$N = kT$$

(Noise power spectral density)

k: Boltzmann Constant k=1.38×10^{-23} J/K

T=Temperature (in Kelvin) [K]=[°C]+273
(for instance 17°C → 290K)

Δf: Bandwidth

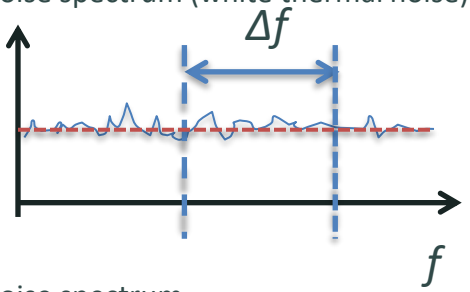


$$v_n = \sqrt{4k \cdot T \cdot R \cdot \Delta f}$$ RMS noise voltage

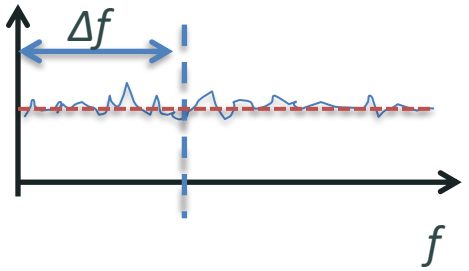
$$P_N = \frac{1}{4}\frac{v_n^2}{R} = k \cdot T \cdot \Delta f$$

Available Noise Power: Maximum noise power delivered to a load

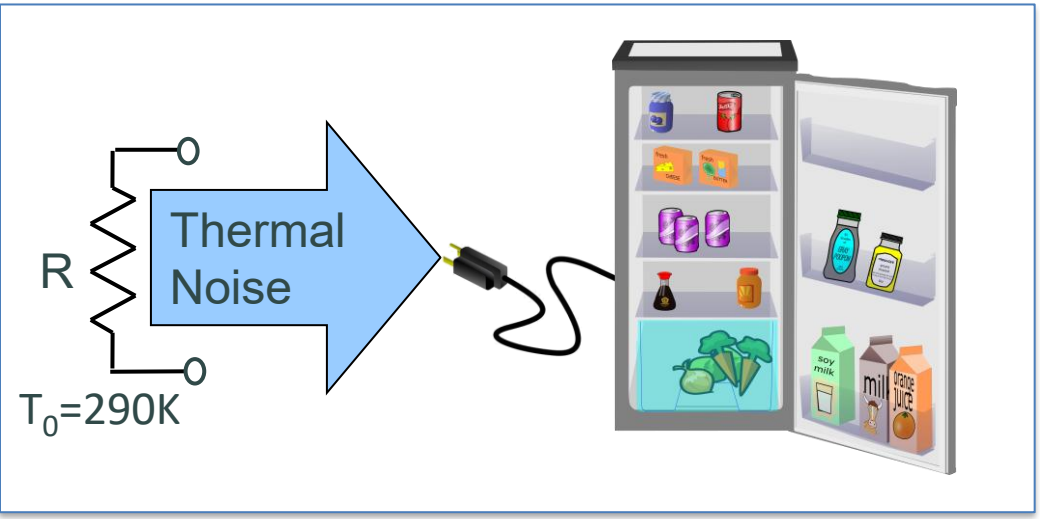
Noise spectrum (white thermal noise)



Noise spectrum



Question

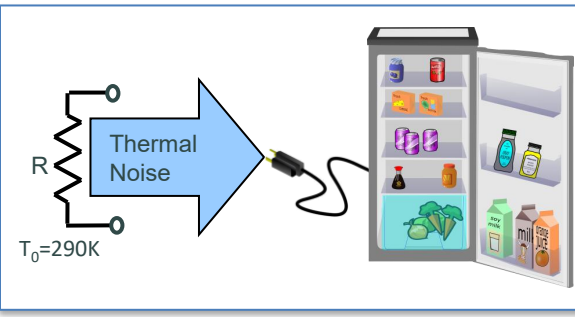


What bandwidth of noise at T_0=290K is needed to power a fridge (400W)?

[k=1.38 10^{-23} J/K]

$P_n = k \cdot T_0 \cdot \Delta f$

Question

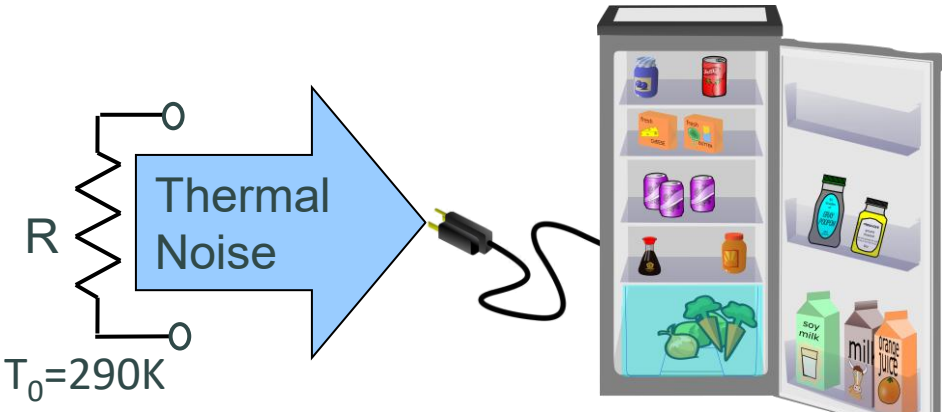


What bandwidth of noise at T_0=290K is needed to power a fridge (400W)?

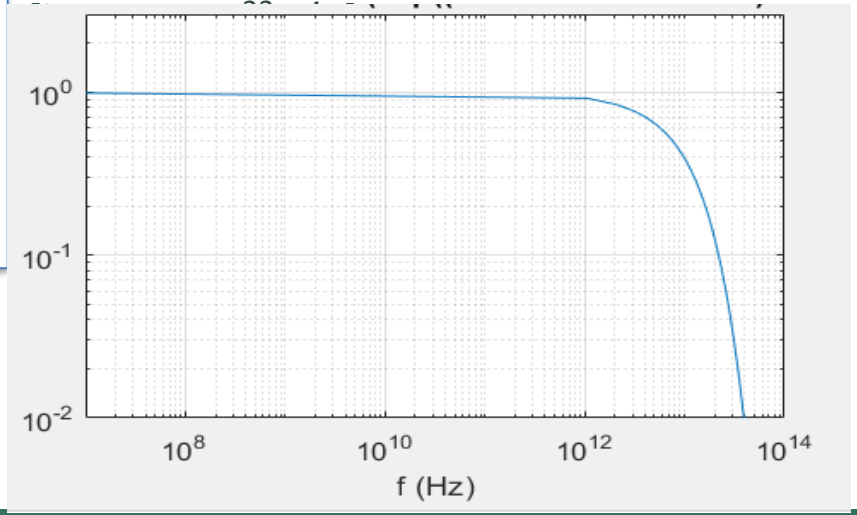
[k=1.38 10^{-23} J/K]

$$P_n = k \cdot T_0 \cdot \Delta f$$

Question



What bandwidth of noise at T_0=290K is needed to power a fridge (400W)?



Using kTΔf → Δf=100ZHz (=10^{14} GHz)

Actually th. noise power density starts decreasing above 1THz→ No appreciable power comes "for free"

→ **Thermal noise is enough only to disturb!**

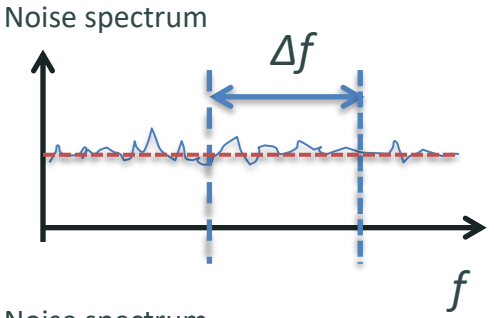
Characterization of Noisy 2-ports

Available noise power at the output of a noiseless two-port:



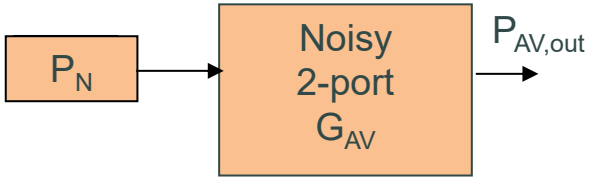
$$P_{AV,out}=G_{AV}\cdot P_{AV,in}=G_{AV}\cdot P_N=G_{AV}\cdot k\cdot T\cdot \Delta f$$

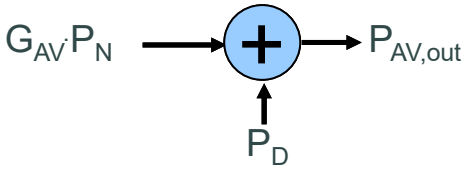
G_{AV}=Available Power Gain

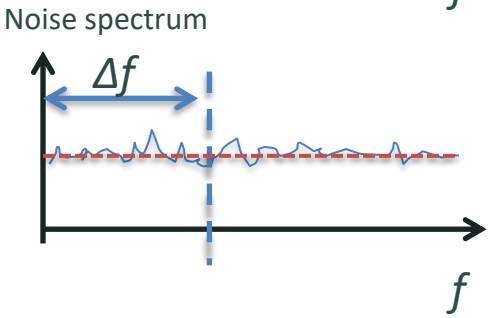


If the 2-port produces noise (P_D), the output power becomes

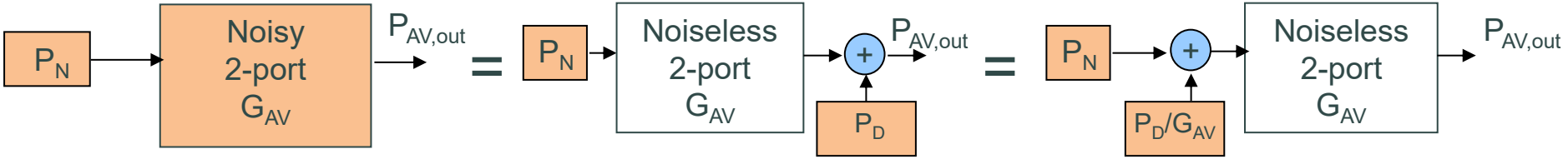
$$P_{AV,out}= G_{AV}P_N + P_D=k\cdot T\cdot \Delta f\cdot G_{AV}+P_D$$







Equivalent Noise Temperature at input

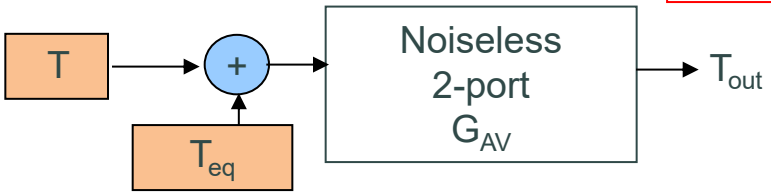


$$P_{AV,out} = k \cdot T \Delta f \cdot G_{AV} + P_D$$

$$P_{AV,out} = k \cdot T \Delta f \cdot G_{AV} + P_D$$

T_{eq}: Equivalent Noise Temperature at the input of the 2-port

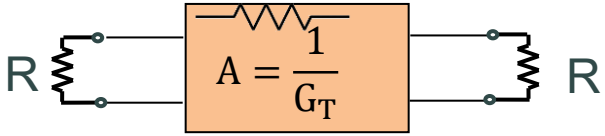
$$P_D / G_{AV} = k \cdot T_{eq} \cdot Df$$



$$P_{AV,out} = k \cdot (T + T_{eq}) \cdot G_{AV} \cdot \Delta f = k \cdot T_{out} \cdot \Delta f$$

$$T_{out} = (T + T_{eq}) \cdot G_{AV}$$

T_{eq} of a Dissipative Attenuator



The 2-port is matched at input and output:

$$G_{AV}=G_T<1.$$

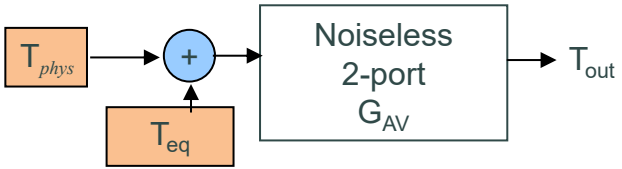
The transducer attenuation A is used: $A=1/G_T>1$

Assuming T_{phys} the physical temperature (Kelvin) of all components:

noise power in the resistor at the output port is

$$P_{out}=k\cdot T_{phys}\cdot \Delta f$$

For thermal equilibrium, it must be equal to the available power at the output port obtained with the eq. model:

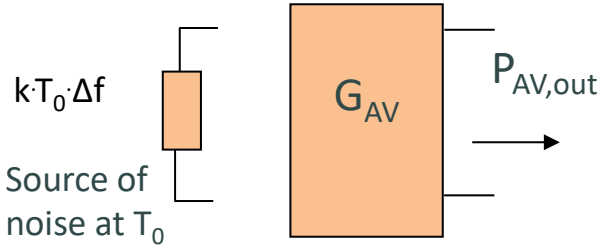


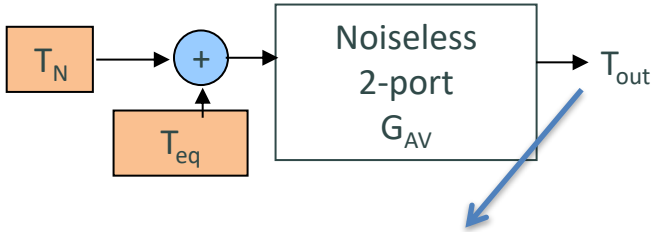
$$P_{AV,out}=k\cdot(T_{phys}+T_{eq})\cdot \Delta f\cdot G_T$$

The equivalent temperature T_{eq} is then given by:

$$T_{eq}=(1/G_T-1)T_{phys}=(A-1)T_{phys}$$

Noise Figure NF of a 2-port





Added noise at output:

$$P_D = G_{AV} k\, T_{eq}\, \Delta f$$

NF is the degradation of the S/N ratio from the input to the output of a 2-port when input is at T_0=290K (input noise $kT_0\Delta f$)

Input signal: P_S

Input noise: $kT_0\Delta f$

Input SNR= $P_S/(kT_0\Delta f)$

Output signal: $G_{AV} P_S$

Output noise: $G_{AV}kT_0\Delta f+P_D$

Output SNR= $G_{AV}P_S/(G_{AV}kT_0\Delta f+P_D)$

$$NF = \frac{SNR_{IN}}{SNR_{OUT}} = \frac{G_{AV} k\, T_0\, \Delta f + P_D}{G_{AV} k\, T_0 \Delta f} = 1 + \frac{P_D}{G_{AV} k\, T_0 \Delta f} = 1 + \frac{T_{eq}}{T_0}$$

T_{eq} as function of Noise Figure NF

$$NF = 1 + \frac{T_{eq}}{T_0}$$ (T_0 = 290 K)

NF is an alternative expression of the equivalent noise temperature at the input of a device, easier to measure than T_{eq}

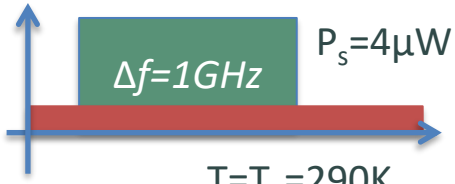
When doing noise computations:

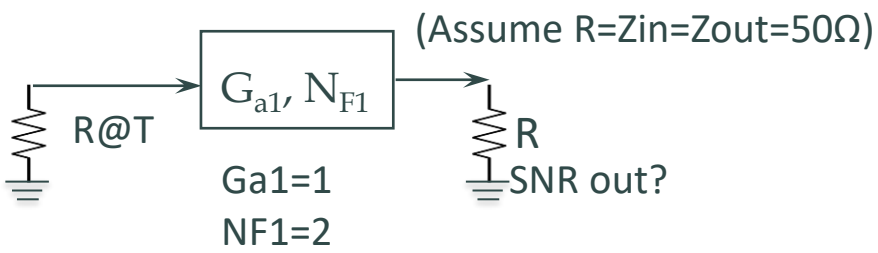
$$T_{eq} = (NF - 1)T_0$$ (T_0 = 290 K)

1. Convert all NF to T_{eq}
2. Convert all other components to their T_{eq}
3. If necessary, draw the noise diagram, replacing every component with its equivalent noise model
4. Do noise computations with temperatures

Noise temperatures can be easily treated as powers and added as independent signals (noise is in fact an additive phenomenon)

Simple example





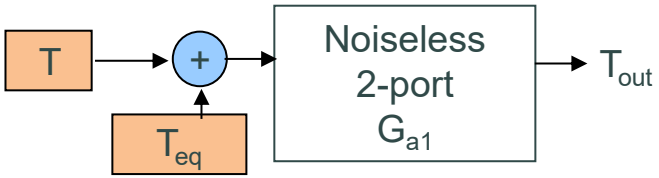
$SNR_{in}=1\cdot10^6$
$T_{eq}=T_0(NF-1)=290K$
$P_{s,out}=P_S\cdot G_{a1}=4\mu W$

$P_D=kT_{eq}\Delta f\, G_{a1}=4pW$
$P_{n,out}=P_nG_{a1}+P_D=8pW$

$T_{out}=(T+T_{eq})G_{a1}=580K$
$P_{n,out}=kT_{out}\,\Delta f=8pW$

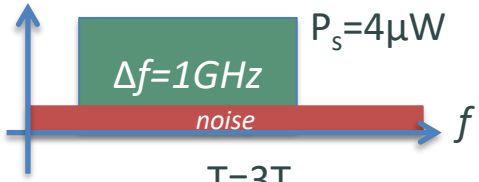
$SNR_{out}=0.5\cdot10^6$





Alternatively, **since the input is at T_0**: $SNR_{out}=SNR_{in}/NF=0.5\cdot10^6$

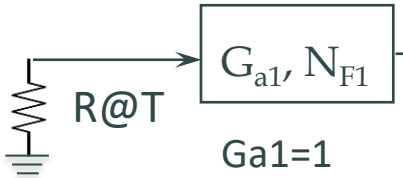
Simple example



P_s=4μW

$T=3T_0$

$P_n=kT\Delta f$=12pW



(Assume $R=Z_{in}=Z_{out}=50\Omega$)

SNR out?

Ga1=1

NF1=2

SNR_{in}=3.3e5

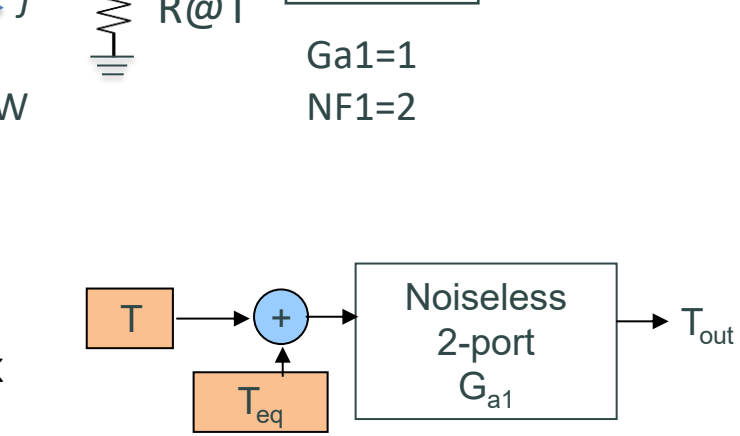
$T_{eq}=T_0(NF-1)=T_0$

$P_{s,out}=P_S \cdot G_{a1}$=4μW

$T_{out}=(T+T_{eq})G_{a1}$=1160K

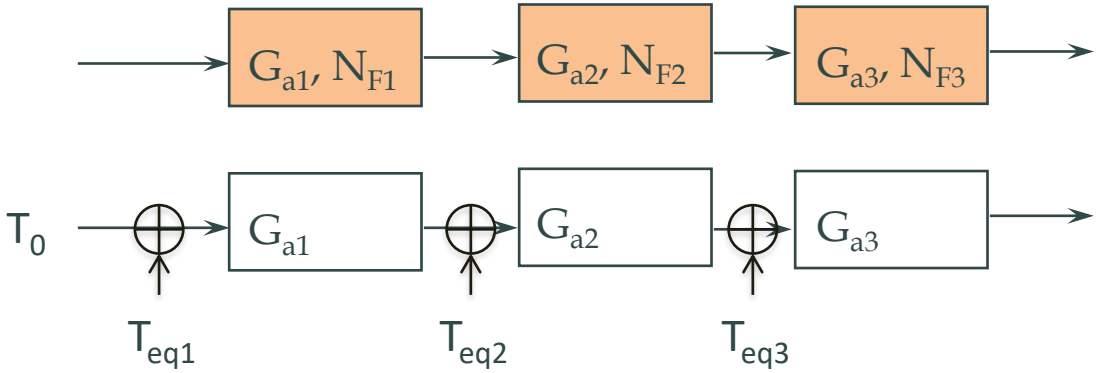
$P_D=kT_{out}\Delta f$=16pW

SNR_{out}=250000



$SNR_{out}=SNR_{in}/NF$=1.7e5 is not correct here (NF is SNR degradation only if T at input =290K)

Noise Figure for cascaded 2-ports

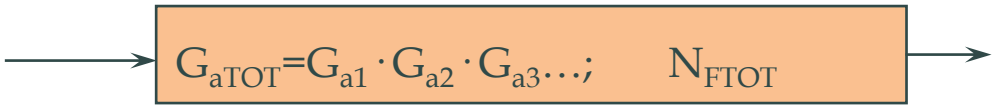


$$Pn_{noiseless} = kT_0\Delta f G_{a3}G_{a2}G_{a1}$$

$$\begin{aligned} Pn_{out} = \; & kT_{eq3}\Delta f G_{a3} \\ & + kT_{eq2}\Delta f G_{a3}G_{a2} \\ & + kT_{eq1}\Delta f G_{a3}G_{a2}G_{a1} \\ & + kT_0\Delta f G_{a3}G_{a2}G_{a1} \end{aligned}$$

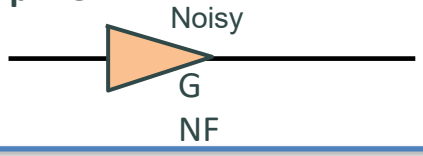
$$(NF)_{TOT} = NF_1 + \frac{NF_2 - 1}{G_{a1}} + \frac{NF_3 - 1}{G_{a1}G_{a2}} + \cdots$$

The equivalent noise figure is dominated by the first stage!



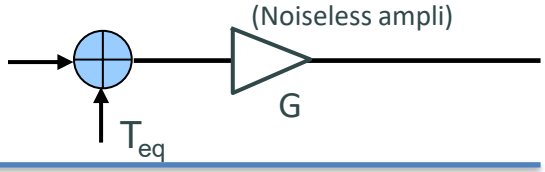
Noise model of some RF components (assumed matched)

General amplifier:

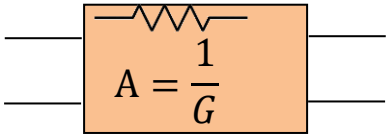


$$T_{eq} = (NF - 1)T_0$$

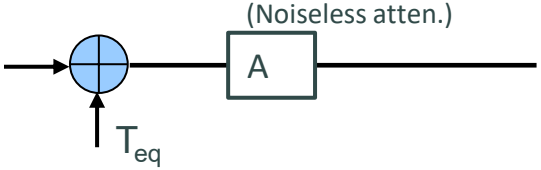
With T_0=290K by definition



Dissipative attenuator at physical temperature T_{phys}:



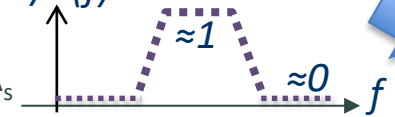
$$T_{eq} = (A - 1)T_{phys}$$



(The NF of a dissipative attenuator at T_{phys}=290K coincides with its attenuation: NF=1+T_{eq}/T_0=A)

Filter at physical temperature T_{phys}:

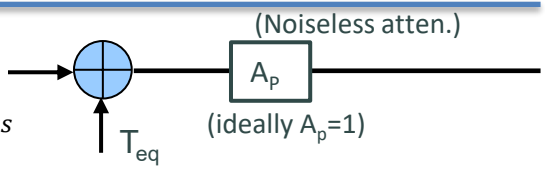




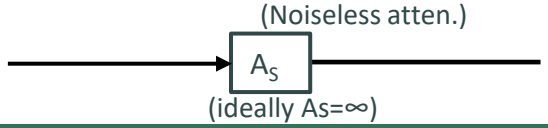
Passband atten. A_p
Stopband rejection A_s

In passband: attenuator!

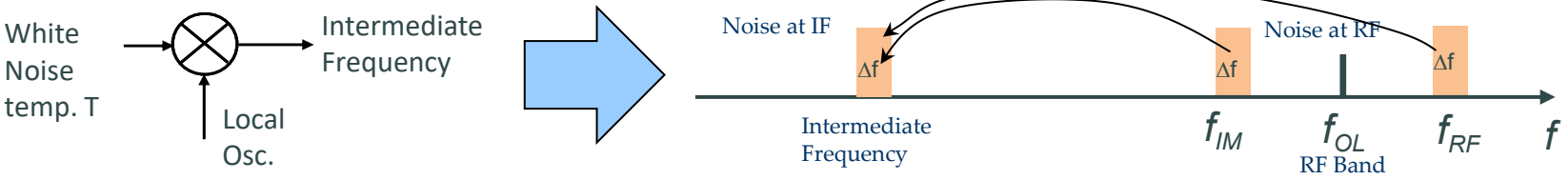
$$T_{eq} = (A_p - 1)T_{phys}$$



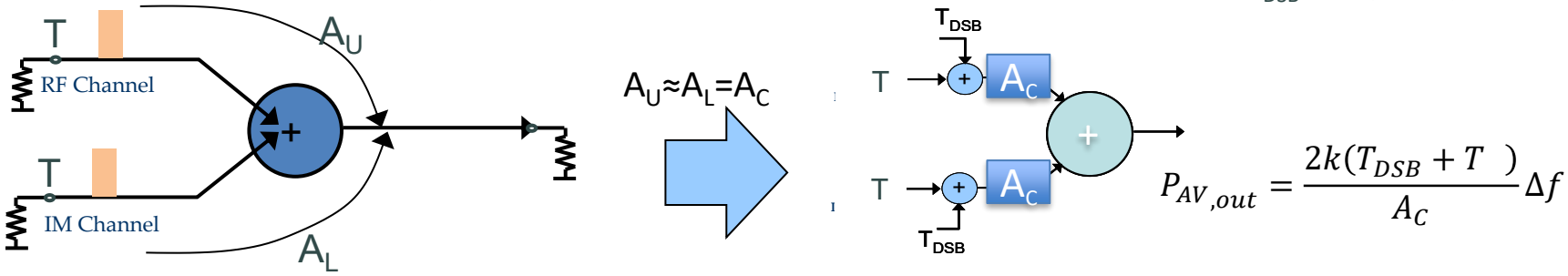
In stopband: rejects by reflection!
(no additional noise)



Noise in Downconversion



The mixer can be considered a 3-port combiner (with attenuations): each branch introduces T_{DSB} as eq. temp.

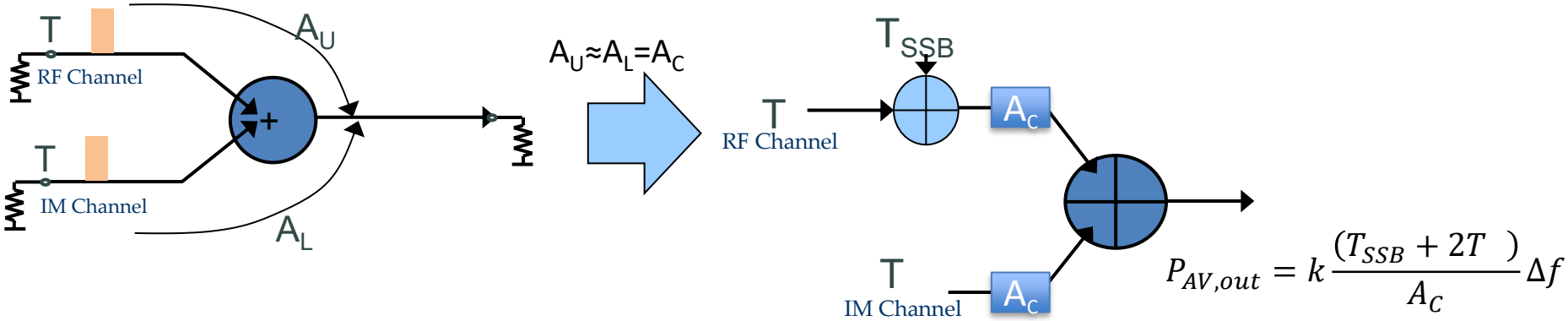


(A real mixer is NOT a dissipative attenuator)

T_{DSB} is usually provided and can be used directly in the noise model

Alternative: SSB Noise Model

Only one equivalent noise source at input:



$$T_{SSB} = 2T_{DSB}$$

T_{SSB} is usually provided and can be used directly in the noise model.

Noise Figure of a Mixer

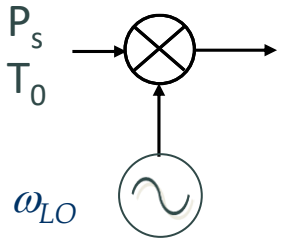
NF: degradation of the SNR from input to output when input is T_0 noise in both bands

P_S : power of the desired RF signal at the input;

T_{SSB} the equivalent noise temperature of the mixer (SSB);

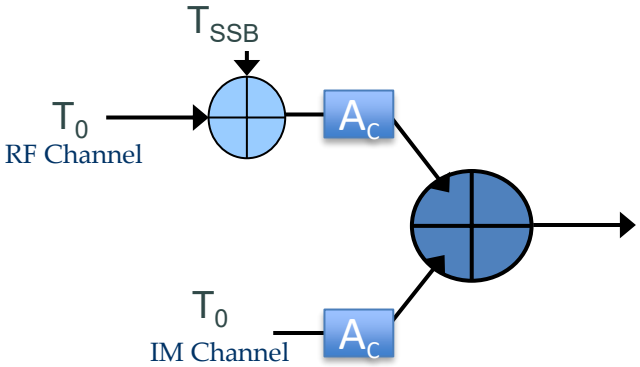
$kT_0\,\Delta f$ the noise power in the RF channel at mixer input

$kT_0\,\Delta f$ the noise power in IM band at mixer input



$$\left(\frac{S}{N}\right)_{in} = \frac{P_S}{kT_0\Delta f}, \qquad \left(\frac{S}{N}\right)_{out} = \frac{\frac{P_S}{A_c}}{\frac{k(2T_0 + T_{SSB})\Delta f}{A_c}}$$

$$NF = \frac{\left(\frac{S}{N}\right)_{in}}{\left(\frac{S}{N}\right)_{out}} = 2 + \frac{T_{SSB}}{T_0}$$



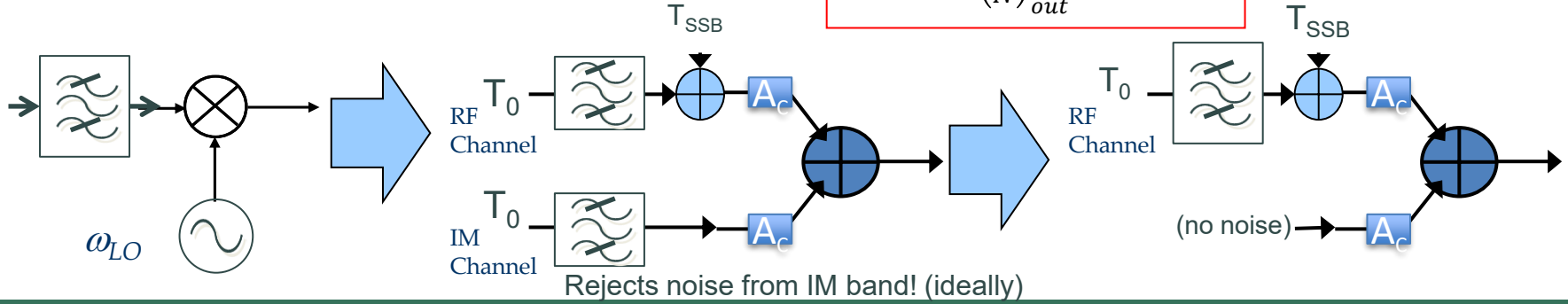
IEEE Noise Figure Definition for a Mixer

- With the previous definition NF=2+T_{SSB}/T_0: a noiseless mixer has NF=2: sounds strange (a lossless 2-port device usually has NF=1)!
- This is due to the inclusion of noise from the image channel in the mixer noise at output
- In practice the mixer is normally preceded by a sharp image filter which rejects the image band (and also the noise in the image band)
- The noise figure thus becomes:

$$\left(\frac{S}{N}\right)_{in} = \frac{P_S}{KT_0\Delta F}, \qquad \left(\frac{S}{N}\right)_{out} = \frac{\frac{P_S}{A_c}}{\frac{K(T_0 + T_{SSB})\Delta F}{A_c}}$$

IEEE Definition (normally used by datasheets)

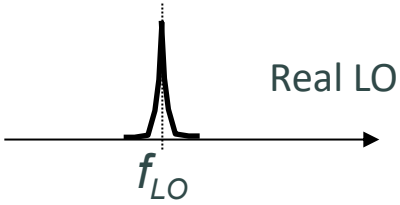
$$NF_{IEEE} = \frac{\left(\frac{S}{N}\right)_{in}}{\left(\frac{S}{N}\right)_{out}} = 1 + \frac{T_{SSB}}{T_0}$$



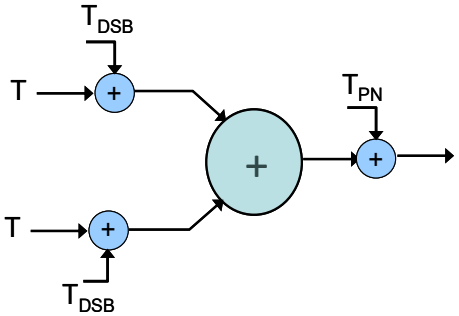
Noise from Local Oscillator

Spectrum from ideal oscillator: a spike at f_{LO} (a single sinusoid at f_{LO}).
In practice: the tone spreads out in a “tent” shape due to the inherent “phase noise” of the LO





In the mixer, the phase noise spectrum is converted to IF as if coming from the RF band.
→ Contributes to worsen the overall noise of the receiver!
→ Represented as an equivalent temperature (T_{PN}) at the mixer output

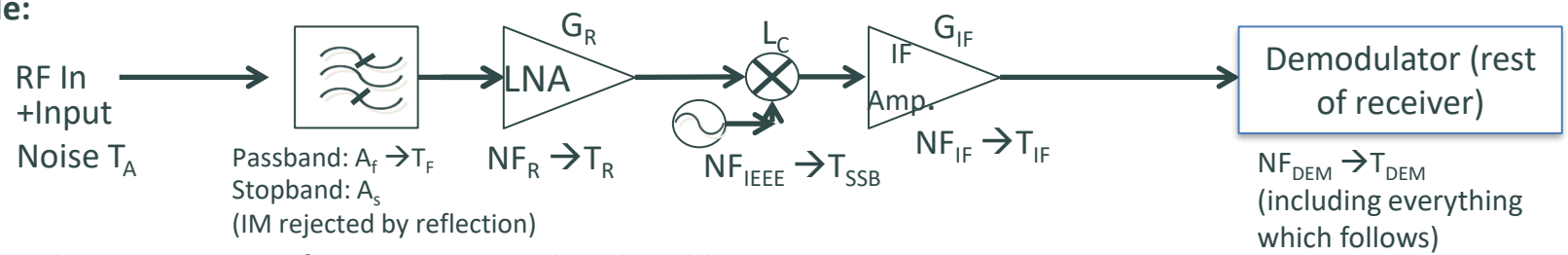


Equivalent Noise and Noise Diagram (IM filter with ∞ rej.)

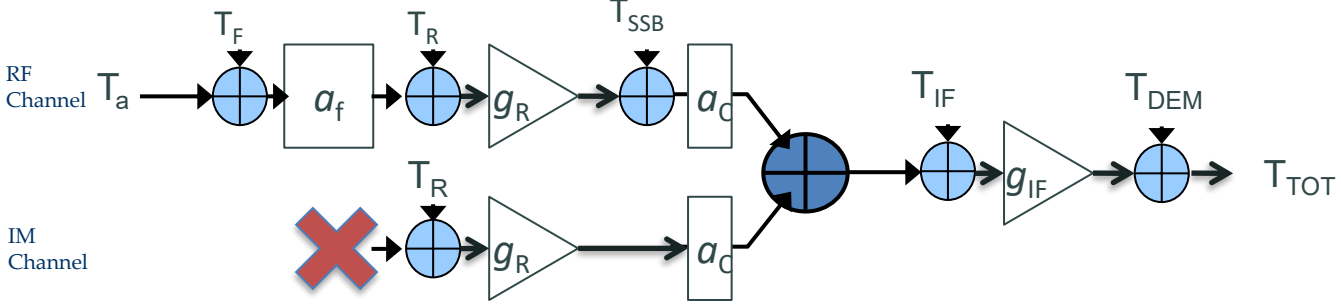
The **equivalent noise temperature of a cascade of blocks** is the equiv. noise (due to all those blocks and only those blocks) at a section (by default at the input, unless otherwise specified)

The **system noise temperature** is the total equiv. noise at a section (usually the input)

Example:

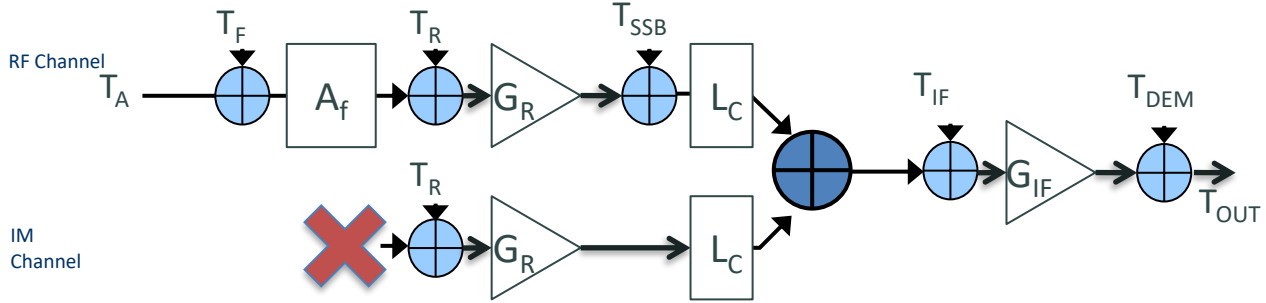


The noise diagram is a step for most noise-related problems:

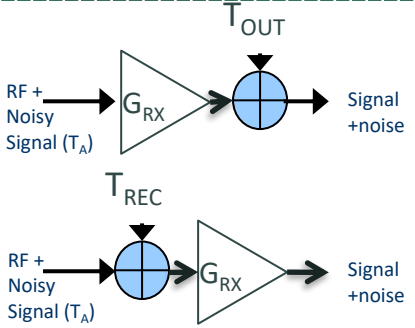


If the IM filter has ∞ rejection in IM band (As=∞dB), what comes before is suppressed!

Example: Equivalent Noise of a Conversion Receiver (IM filter with ∞ rej.)



Equiv. to



Example: Total noise at output due to the receiver alone (→neglect T_A of input signal):

$$T_{out} = T_{DEM} + T_{IF}G_{IF} + T_{SSB}\frac{G_{IF}}{L_C} + T_R\frac{G_R G_{IF}}{L_C} + T_F\frac{G_R G_{IF}}{A_f L_C} + T_R\frac{G_R G_{IF}}{L_C}$$

Example: equivalent noise at input due to the receiver alone (→neglect T_A of input signal)

$$T_{REC} = T_F + A_f T_R + \frac{A_f}{G_R}T_{SSB} + T_R\frac{G_R}{L_C}\frac{L_C}{G_R}A_f + T_{IF}\frac{L_C}{G_R}A_f + T_{DEM}\frac{1}{G_{IF}}\frac{L_C}{G_R}A_f$$

Example: system noise at input (include T_A of input signal)

$$T_{sys} = T_A + T_F + A_f T_R + \frac{A_f}{G_R}T_{SSB} + T_R\frac{G_R}{L_C}\frac{L_C}{G_R}A_f + T_{IF}\frac{L_C}{G_R}A_f + T_{DEM}\frac{1}{G_{IF}}\frac{L_C}{G_R}A_f$$

$$G_{RX} = \frac{1}{A_f}G_R\frac{1}{L_C}G_{IF}$$

Relation between the two:

$$T_{REC} = \frac{T_{out}}{G_{RX}}$$

The LNA here is not followed by a filter
→ its noise is injected in the image band too
→ T_R appears twice